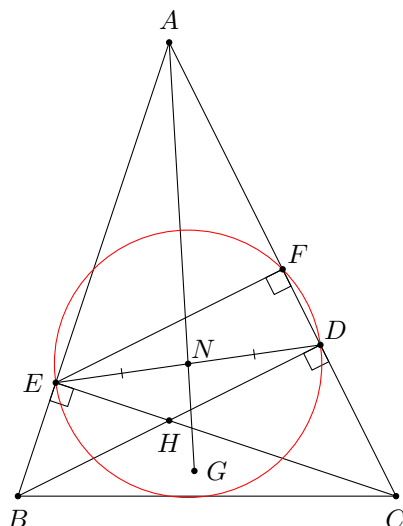


Problem 9 (Sharygin Geometry Olympiad Final 2015 Grade 9 Problem 7). Let ABC be an acute, non-isosceles triangle. The altitudes BD and CE intersect at H . Let G be the centroid of $\triangle HBC$. If the line AG bisects the segment DE , find $\angle A$.



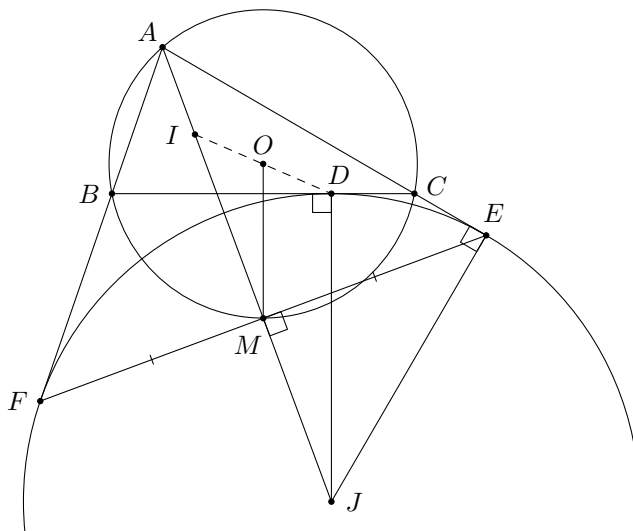
Solution. Note that A is the orthocentre of $\triangle HBC$. So AG is the Euler line of $\triangle HBC$. Let N be the nine-point centre of $\triangle HBC$. Then N lies on the Euler line AG and $ND = NE$ since D and E are the feet of altitudes. Since $\triangle ABC$ is non-isosceles, we have $AD \neq AE$, which means AG cannot be the perpendicular bisector of DE . Therefore, there is only one point on AG which is equidistant from D and E . By assumption, this is the intersection of AG and DE . This means N is this intersection point.

Let F be the midpoint of AC , which lies on the nine-point circle of $\triangle HBC$. Since N lies on DE , DE is a diameter of the nine-point circle. So $\angle EFD = 90^\circ$. As $AF = FC$, this means EF is the perpendicular bisector of AC . Thus,

$$\angle EAC = \angle ECA = \frac{90^\circ}{2} = 45^\circ.$$

This means $\angle BAC = 45^\circ$. □

Problem 10 (All-Russian Olympiad 2003 Grade 10 Problem 7). Let O and I be the circumcentre and incentre of $\triangle ABC$ respectively. The A -excircle of $\triangle ABC$ touches the lines BC, CA, AB at D, E, F respectively. If the midpoint of EF lies on the circumcircle of $\triangle ABC$, prove that O, I, D are collinear.



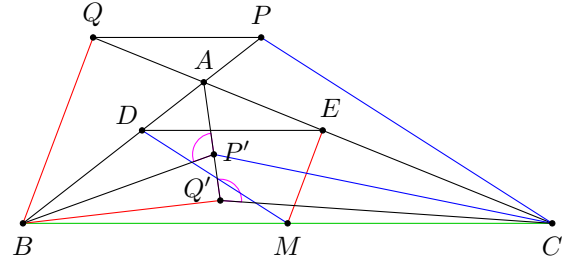
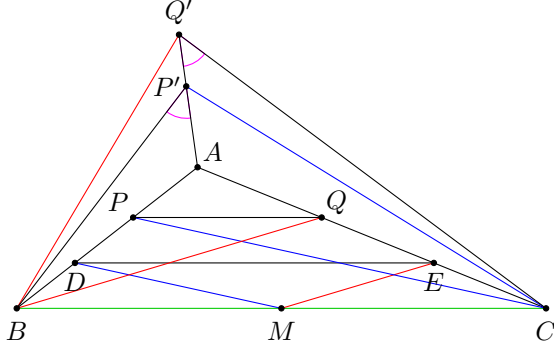
Solution. Let J be the A -excentre, and let M be the midpoint of EF . Since $AE = AF$, M lies on the internal angle bisector AIJ . As M lies on the circumcircle, M is the midpoint of IJ . Note that both MO and JD are perpendicular to BC , hence they are parallel. Therefore, it suffices to show $JD = 2MO$, from which it follows that $\triangle IMO \sim \triangle IJD$, which implies I, O, D are collinear.

Let R and r_a be the circumradius and A -exradius of $\triangle ABC$. Then we need to prove $r_a = 2R$. Now, as $\angle JME = \angle AEJ = 90^\circ$, we have $\triangle EMJ \sim \triangle AEJ$. Thus,

$$r_a^2 = JE^2 = JM \times JA = \text{power of } J \text{ with respect to } (ABC) = OJ^2 - R^2 = 2Rr_a.$$

The last equality is analogous to Euler's formula for OI^2 . This yields $r_a = 2R$ as desired. $\square$

Problem 11 (239 Open MO 2008 Grade 10-11 Problem 5). In $\triangle ABC$, $\angle A = 120^\circ$. Let M be the midpoint of BC . Points D and E are chosen on the sides AB and AC respectively, such that $DE \parallel BC$. Prove that $MD + ME \geq MB$.



Solution 1. Reflect B and C about D and E respectively to obtain points P and Q . Then reflect P and Q about AC and AB respectively to obtain points P' and Q' . By the midpoint theorem, we have $MD = \frac{1}{2}CP = \frac{1}{2}CP'$ and similarly $ME = \frac{1}{2}BQ'$. So it remains to prove $CP' + BQ' \geq BC$.

Since $\angle BAC = 120^\circ$ and P lies on AB , the point P' lies on the internal angle bisector of $\angle BAC$. Similarly, Q' lies on the same angle bisector. WLOG assume P' lies between A and Q' (possibly $P' = Q'$). Also, from $DE \parallel BC$, we have $PQ \parallel BC$ and hence $\frac{AP}{AQ} = \frac{AB}{AC}$. This yields $\frac{AP'}{AB} = \frac{AQ'}{AC}$. Together with $\angle BAP' = \angle CAQ'$, we have $\triangle ABP' \sim \triangle ACQ'$. Let $\angle AP'B = \angle AQ'C = \theta$.

- Case 1: P and Q lie on the closed segments AB and AC respectively.

In this case, P' and Q' lie outside $\triangle ABC$ and lie on the same side of BC as A . So $\angle BP'Q' > \angle BAP' = 120^\circ$. This shows $\angle BP'Q'$ is the largest interior angle of $\triangle BP'Q'$, hence $CP' + BQ' \geq CP' + BP' \geq BC$ by the triangle inequality.

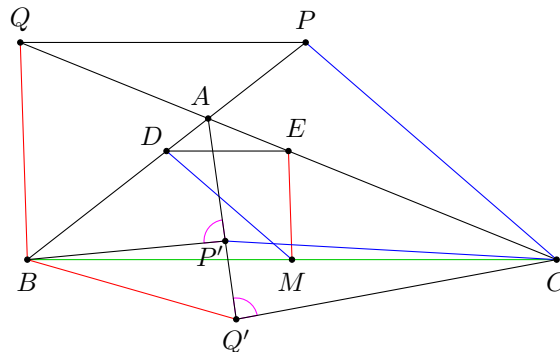
- Case 2: P and Q lie outside $\triangle ABC$, and $\theta \geq 90^\circ$.

In this case, $\angle CQ'P' = \theta \geq 90^\circ$ is the largest interior angle of $\triangle CP'Q'$. Thus, $CP' + BQ' \geq CQ' + BQ' \geq BC$. Equality holds when $P' = Q'$ and Q' lies on BC , i.e. when $AB = AC$ and D, E are the projection of M onto AB, AC .

- Case 3: P and Q lie outside $\triangle ABC$, and $\theta < 90^\circ$.

In this case, $\angle BP'Q' = 180^\circ - \theta > 90^\circ$ is the largest interior angle of $\triangle BP'Q'$. Thus, $CP' + BQ' \geq CP' + BP' \geq BC$.

So we are done.



$\square$

Solution 2. Denote $a = BC$, $b = CA$, $c = AB$ as usual. Let $AD = kAB$ so that $AE = kAC$. Applying Stewart's theorem to $\triangle ABM$, we have

$$MD^2 = \frac{1}{AB}(AD \cdot BM^2 + BD \cdot AM^2) - AD \cdot BD = \frac{1}{c} \left(\frac{ka^2c}{4} + \frac{(1-k)c(2b^2 + 2c^2 - a^2)}{4} \right) - k(1-k)c^2.$$

Using the cosine formula, we have $a^2 = b^2 + c^2 + bc$. After simplification, we obtain $MD^2 = \frac{b^2 - (1-2k)bc + (1-2k)^2c^2}{4}$. Let $t = 1 - 2k$. Then $2MD = \sqrt{b^2 - tbc + t^2c^2}$. Similarly, $2ME = \sqrt{c^2 - tbc + t^2b^2}$. It remains to prove

$$\sqrt{b^2 - tbc + t^2c^2} + \sqrt{c^2 - tbc + t^2b^2} \geq \sqrt{b^2 + bc + c^2}.$$

(This is essentially $BQ + CP \geq BC$ as in solution 1. Here is an algebraic proof.) Squaring both sides and simplifying, we need to prove

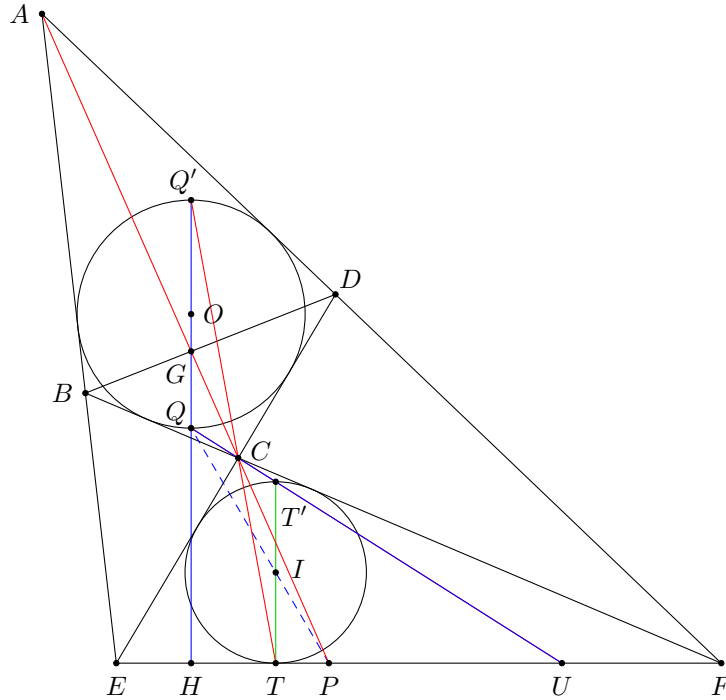
$$2\sqrt{(b^2 - tbc + t^2c^2)(c^2 - tbc + t^2b^2)} \geq (2t+1)bc - t^2(b^2 + c^2).$$

Note that $b^2 - tbc + t^2c^2 = \left(tc - \frac{b}{2}\right)^2 + \frac{3b^2}{4} \geq \frac{3b^2}{4}$, and similarly we have $c^2 - tbc + t^2b^2 \geq \frac{3c^2}{4}$. Thus, it suffices to prove $\frac{3}{2}bc \geq (2t+1)bc - t^2(b^2 + c^2)$. This is true since

$$t^2(b^2 + c^2) \geq 2t^2bc \geq \left(2t - \frac{1}{2}\right)bc.$$

Equality holds when $t = \frac{1}{2}$ and $b = c$. □

Problem 12 (Brazil MO 2017 Problem 3). Let ω be the incircle of a quadrilateral $ABCD$. The rays AB and DC meet at point E . The rays AD and BC meet at point F . The lines AC and EF intersect at point P . Let Q be the point on ω which is closest to the line EF . Prove that the line PQ passes through the incentre I of $\triangle CEF$.



Solution. Define the following.

- O is the centre of ω
- T is the contact point of the incircle of $\triangle CEF$ with EF
- T' is the point on the incircle of $\triangle CEF$ such that TT' is a diameter
- Q' is the point on ω such that QQ' is a diameter
- G is the intersection point of AC and BD
- H is the foot of perpendicular from Q to EF
- U is the intersection of QC and EF

We have the following facts.

- $IT \parallel OQ$ (since both are perpendicular to EF)
- Q, C, T' are collinear (by homothety since C is the internal centre of similitude of ω and the incircle of $\triangle CEF$)
- Q', C, T are collinear (by the same homothety)
- G lies on OQH (since $OG \perp EF$ by Brokard's theorem)
- $(H, G; Q, Q')$ is a harmonic division (since EF is indeed the polar line of G with respect to ω)

Now, we have the following projection: $HQ(H, G, Q, Q') \xrightarrow{C} EF(H, P, U, T) \xrightarrow{Q} TI(\infty_{TI}, TI \cap PQ, T', T)$. This shows $(\infty_{TI}, TI \cap PQ; T', T)$ is a harmonic division. Thus, $TI \cap PQ$ is the midpoint of $T'T$, i.e. $TI \cap PQ = I$. This means PQ passes through I . $\square$